

# Zhe Chen 陈哲

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github.com/CZ114

**Personal website:** <https://cz114.github.io/>



## Education

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| <b>Imperial College London</b>                                                  | 2025.10 – 2026.09 |
| MRes, Medical Robotics   Research completed; formal degree award pending        |                   |
| <b>University of Nottingham Ningbo China</b>                                    | 2021.09 – 2025.06 |
| BEng (Hons), Electrical and Electronic Engineering   <b>First-Class Honours</b> |                   |

## Technical Skills

**Agents and Evaluation:** Function Calling, RAG, workflows and memory; BFCL calibration, retrieval evaluation, LOSO, ablation studies and evidence traceability.

**Algorithm Foundations:** Python, PyTorch, A2C/DQN, federated learning and pseudo-labelling; systematic study of Transformer/Attention and the principles of SFT, RLHF and PPO.

**Robotics and Embedded Systems:** ESP32-S3, I2S, BLE, UDP, MQTT, CoppeliaSim, robot kinematics, Ubuntu, Jetson Nano and Altium Designer.

## Research Experience

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| <b>Imperial College London • Hamlyn Centre • I-thread Lab</b>                        | Current |
| Research Assistant (RA)   AD-related wearable sensing, AI agents and speech research |         |

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|------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Speech Evidence &amp; Clinician Report Chat</b>                                                                                       | 2026 |
| Imperial College London • Hamlyn Centre • I-thread Lab   Research Support<br>Manuscript in preparation • Target journal: Nature Medicine |      |

- Assisted with checking and verifying speech metrics within the collaborative research project.
- Helped organise the report page and its existing content for review.
- Implemented part of the chat functionality within the report interface.

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| <b>UNNC Smart Medicine Laboratory</b>                                                   | 2023.09 – 2025.07 |
| Research Intern / Project Lead   Smart-Home IoT and Federated Learning for Sleep Apnoea |                   |

- Addressed the lack of expert labels for home heart-rate data. Built an M5Stack/ESP32 + MQTT gateway, configured FATE/PyTorch experiments, and ran **confidence-thresholded pseudo-labelling and four federated training rounds**, comparing pseudo-labels with a control treating home samples as normal.
- Achieved **81% external-test accuracy in round four versus 76% for the control (+5 percentage points)**; sleep-apnoea class precision / recall were **0.93 / 0.78**, supporting the feasibility of improving models with unlabelled home data.
- Contributed to drafting a collaborative manuscript as a listed co-author, providing experimental evidence for collaborative modelling with home data lacking expert annotations.

## Projects

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### Multimodal AI Analysis Platform for Alzheimer's Disease Research

2026.01 – 2026.09

Imperial College London | Hamlyn Centre · I-thread Lab | MRes Research

Poster exhibited at the Hamlyn Symposium on Medical Robotics 2026.

Developed a **wearable-to-agent research platform** for multimodal monitoring, integrating ESP32 sensing, signal processing, retrieval and agent workflows into evidence-linked reports.

- **Multimodal sensing:** Integrated PPG, GSR, IMU and audio on ESP32-S3; sustained approximately **50 Hz** acquisition with no detected frame loss across **136,562 frames / 27 serial recordings**.
- **Heart-rate estimation:** Combined spectral analysis, autocorrelation and motion suppression; achieved **MAE 5.57 bpm vs 11.84 bpm** for the device baseline (PPG-DaLiA, 15 participants, LOSO).
- **Agent workflows:** Built deterministic tools, retrieval, memory and bounded workflows; improved movement accuracy from **35.7% to 81.4%** in a fixed-model ablation, with parsing failures reduced from **15 to 0**.
- **Movement recognition:** Used 18-feature inputs for 5-class recognition; achieved **82.9% accuracy / 0.804 macro-F1**, versus **78.6% / 0.713** for random forest (70 recordings, 6 healthy volunteers, LOSO).
- **RAG and retrieval:** Implemented hybrid retrieval and HyDE comparisons; achieved **Recall@10 = 0.846** and **MRR = 0.753–0.774** on a 24-question internal set.
- **Tool-use and reporting:** Aligned the in-house BFCL scorer with official decisions on **400/400 cases** (both scored **91.25%**); **66/70 (94.3%)** reports met all traceability checks.
- **AgentClinic simulation:** Reduced mean dialogue turns for all six models; two model conditions improved task accuracy from **53% / 77% to 67% / 87%**.

### Reinforcement Learning for Port Container Relocation

2024.08 – 2025.06

University of Nottingham Ningbo China | Undergraduate Dissertation

- Built a three-dimensional container-yard simulator to reduce unnecessary handling; designed state representations and reward functions, and implemented A2C/DQN training and evaluation pipelines.
- Against a **random-action baseline**, DQN reduced rehandles by approximately **23% in simple scenarios**, while A2C reduced handling moves by approximately **40% in complex scenarios**, demonstrating scope for learning-based scheduling improvements.

## Awards

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### Best IoT Project Award – DeskPal, UNNC (EEEE3124)

2024.12

### DeskPal – Embedded AI Companion

2024.12

University of Nottingham Ningbo China | Embedded ML and IoT

- Built a Wio Terminal desktop companion; trained an **Edge Impulse MLP** for three motion classes (idle, rise and wave) and deployed on-device inference.
- Integrated environmental sensing, LCD expression / reward states and **bidirectional MQTT** for interactive feedback, reminders and data exchange.

## Open-Source Tools

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**Vibedog-for-agents** 2026.05 – 2026.06 | AI-assisted development

Agent activity display, human approval and a stop control. [github.com/CZ114/Vibedog-for-agents](https://github.com/CZ114/Vibedog-for-agents)

**esp32-audio-ai** | Audio acquisition and transcription precursor to the main project

ESP32 audio capture, Whisper transcription, and pause / speech-rate statistics.

[github.com/CZ114/esp32-audio-ai](https://github.com/CZ114/esp32-audio-ai)